

● HARD

● ADVANCED MATH

● ANSWER KEY

Advanced Math

12

10 answers · Rationale-rich · Teach from doc

Q1**231**

The correct answer is 231. It's given that $\sqrt{k-x} = 58-x$. Squaring both sides of this equation yields $k-x = (58-x)^2$, which is equivalent to the given equation if $58-x > 0$. It follows that if a solution to the equation $k-x = (58-x)^2$ satisfies $58-x > 0$, then it's also a solution to the given equation; if not, it's extraneous. The equation $k-x = (58-x)^2$ can be rewritten as $k-x = 3,364 - 116x + x^2$. Adding x to both sides of this equation yields $k = x^2 - 115x + 3,364$. Subtracting k from both sides of this equation yields $0 = x^2 - 115x + (3,364 - k)$. The number of solutions to a quadratic equation in the form $0 = ax^2 + bx + c$, where a , b , and c are constants, can be determined by the value of the discriminant, $b^2 - 4ac$. Substituting -115 for b , 1 for a , and $3,364 - k$ for c in $b^2 - 4ac$ yields $(-115)^2 - 4(1)(3,364 - k)$, or $4k - 231$. The equation $0 = x^2 - 115x + (3,364 - k)$ has exactly one real solution if the discriminant is equal to zero, or $4k - 231 = 0$. Subtracting 231 from both sides of this equation yields $4k = 231$. Dividing both sides of this equation by 4 yields $k = 57.75$. Therefore, if $k = 57.75$, then the equation $0 = x^2 - 115x + (3,364 - k)$ has exactly one real solution. Substituting 57.75 for k in this equation yields $0 = x^2 - 115x + (3,364 - 57.75)$, or $0 = x^2 - 115x + 3,306.25$, which is equivalent to $0 = (x - 57.5)^2$. Taking the square root of both sides of this equation yields $0 = x - 57.5$. Adding 57.5 to both sides of this equation yields $57.5 = x$. To check whether this solution satisfies $58 - x > 0$, the solution, 57.5, can be substituted for x in $58 - x > 0$, which yields $58 - 57.5 > 0$, or $0.5 > 0$. Since 0.5 is greater than 0, it follows that if $k = 57.75$, or $4k = 231$, then the given equation has exactly one real solution. If $4k < 231$, then the discriminant, $4k - 231$, is negative and the given equation has no solutions. Therefore, the minimum possible value of $4k$ is 231.

The correct answer is 609. It's given that the function g is defined by $gx = \frac{x}{a} - 14$, where $a < 0$. Substituting $15a$ for x in function g yields $g15a = \frac{15a}{a} - 14$. This function can be rewritten as $g15a = \frac{15a}{a} - 14$, or $g15a = 15\frac{a}{a} - 14$. Since $a < 0$, it follows that $\frac{a}{a} = -1$. Substituting -1 for $\frac{a}{a}$ in $g15a = 15\frac{a}{a} - 14$ yields $g(15a) = 15(-1) - 14$, or $g(15a) = -29$. Similarly, substituting $7a$ for x in function g yields $g7a = \frac{7a}{a} - 14$. This function can be rewritten as $g7a = \frac{7a}{a} - 14$, or $g7a = 7\frac{a}{a} - 14$. Since $a < 0$, it again follows that $\frac{a}{a} = -1$. Substituting -1 for $\frac{a}{a}$ in $g7a = 7\frac{a}{a} - 14$ yields $g(7a) = 7(-1) - 14$, or $g(7a) = -21$. Therefore, $g(15a) = -29$ and $g(7a) = -21$. Thus, the product of $g(15a)$ and $g(7a)$ is $(-29)(-21)$, or 609.

Q3**A****A. -3**

Choice A is correct. The number of solutions to any quadratic equation in the form $ax^2 + bx + c = 0$, where a , b , and c are constants, can be found by evaluating the expression $b^2 - 4ac$, which is called the discriminant. If the value of $b^2 - 4ac$ is a positive number, then there will be exactly two real solutions to the equation. If the value of $b^2 - 4ac$ is zero, then there will be exactly one real solution to the equation. Finally, if the value of $b^2 - 4ac$ is negative, then there will be no real solutions to the equation.

The given equation $2x^2 - 4x = t$ is a quadratic equation in one variable, where t is a constant. Subtracting t from both sides of the equation gives $2x^2 - 4x - t = 0$. In this form, $a = 2$, $b = -4$, and $c = -t$. The values of t for which the equation has no real solutions are the same values of t for which the discriminant of this equation is a negative value. The discriminant is equal to $(-4)^2 - 4(2)(-t)$; therefore, $(-4)^2 - 4(2)(-t) < 0$. Simplifying the left side of the inequality gives $16 + 8t < 0$. Subtracting 16 from both sides of the inequality and then dividing both sides by 8 gives $t < -2$. Of the values given in the options, -3 is the only value that is less than -2 . Therefore, choice A must be the correct answer.

Choices B, C, and D are incorrect and may result from a misconception about how to use the discriminant to determine the number of solutions of a quadratic equation in one variable.

Q4

B

B. -9

Choice B is correct. It's given that $gx = fx - 1$. Since $fx = x + 6x + 5x + 1$, it follows that $fx - 1 = x - 1 + 6x - 1 + 5x - 1 + 1$. Combining like terms yields $fx - 1 = x + 5x + 4x$. Therefore, $gx = xx + 5x + 4$. The x -intercepts of a graph in the xy -plane are the points where $y = 0$. The x -coordinates of the x -intercepts of the graph of $y = gx$ in the xy -plane can be found by solving the equation $0 = xx + 5x + 4$. Applying the zero product property to this equation yields three equations: $x = 0$, $x + 5 = 0$, and $x + 4 = 0$. Solving each of these equations for x yields $x = 0$, $x = -5$, and $x = -4$, respectively. Therefore, the x -intercepts of the graph of $y = gx$ are $0, 0, -5, 0$, and $-4, 0$. It follows that the values of a , b , and c are $0, -5$, and -4 . Thus, the value of $a + b + c$ is $0 + -5 + -4$, which is equal to -9 .

Choice A is incorrect. This is the value of $a + b + c$ if $gx = fx + 1$.

Choice C is incorrect. This is the value of $a + b + c - 1$ if $gx = x - 6x - 5x - 1$.

Choice D is incorrect. This is the value of $a + b + c$ if $fx = x - 6x - 5x - 1$.

Q5**A****A.** $\frac{1}{57}$

Choice A is correct. The left-hand side of the given equation is the expression $57x^2 + 57b + ax + ab$. Applying the distributive property to this expression yields $57x^2 + 57bx + ax + ab$. Since the first two terms of this expression have a common factor of $57x$ and the last two terms of this expression have a common factor of a , this expression can be rewritten as $57xx + b + ax + b$. Since the two terms of this expression have a common factor of $x + b$, it can be rewritten as $x + b57x + a$. Therefore, the given equation can be rewritten as $x + b57x + a = 0$. By the zero product property, it follows that $x + b = 0$ or $57x + a = 0$. Subtracting b from both sides of the equation $x + b = 0$ yields $x = -b$. Subtracting a from both sides of the equation $57x + a = 0$ yields $57x = -a$. Dividing both sides of this equation by 57 yields $x = \frac{-a}{57}$. Therefore, the solutions to the given equation are $-b$ and $\frac{-a}{57}$. It follows that the product of the solutions of the given equation is $-b\frac{-a}{57}$, or $\frac{ab}{57}$. It's given that the product of the solutions of the given equation is kab . It follows that $\frac{ab}{57} = kab$, which can also be written as $ab\frac{1}{57} = abk$. It's given that a and b are positive constants. Therefore, dividing both sides of the equation $ab\frac{1}{57} = abk$ by ab yields $\frac{1}{57} = k$. Thus, the value of k is $\frac{1}{57}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6**D****D. Zero**

Choice D is correct. Each point x, y on the graph of $y = f(x)$ in the xy -plane gives a value of x and its corresponding value of $f(x)$, or y . For any value of x for which $f(x) = 0$, there is a corresponding point on the graph of $y = f(x)$ with a y -coordinate of 0. A point with a y -coordinate of 0 is a point where the graph intersects the x -axis. It's given that $f(x) = ab^x + c$, where a , b , and c are constants. In the xy -plane, the graph of an equation of this form will lie entirely either above or below the horizontal line defined by $y = c$. The part of the graph of $y = f(x)$ shown lies entirely below the horizontal line defined by $y = -7$, and thus the entire graph of $y = f(x)$ must lie below the line defined by $y = -7$. It follows that the graph of $y = f(x)$ will never intersect the x -axis. Therefore, there are zero values of x for which $f(x) = 0$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Q7**.0625**

Accept also: 1/16

The correct answer is $\frac{1}{16}$. Let p and q represent the solutions to the given equation. Then, the given equation can be rewritten as $64x - px - q = 0$, or $64x^2 - 64p + q + pq = 0$. Since this equation is equivalent to the given equation, it follows that $-16a + 4b = -64p + q$. Dividing both sides of this equation by -64 yields $\frac{16a + 4b}{64} = p + q$, or $\frac{1}{16}4a + b = p + q$. Therefore, the sum of the solutions to the given equation, $p + q$, is equal to $\frac{1}{16}4a + b$. Since it's given that the sum of the solutions to the given equation is $k4a + b$, where k is a constant, it follows that $k = \frac{1}{16}$. Note that $1/16$, $.0625$, 0.062 , and 0.063 are examples of ways to enter a correct answer.

Alternate approach: The given equation can be rewritten as $64x^2 - 44a + bx + ab = 0$, where a and b are positive constants. Dividing both sides of this equation by 4 yields

$16x^2 - 4a + bx + \frac{ab}{4} = 0$. The solutions for a quadratic equation in the form $Ax^2 + Bx + C = 0$, where A , B , and C are constants, can be calculated using the quadratic formula,

$x = \frac{-B + \sqrt{B^2 - 4AC}}{2A}$ and $x = \frac{-B - \sqrt{B^2 - 4AC}}{2A}$. It follows that the sum of the solutions to a quadratic equation in the form $Ax^2 + Bx + C = 0$ is $\frac{-B + \sqrt{B^2 - 4AC}}{2A} + \frac{-B - \sqrt{B^2 - 4AC}}{2A}$, which can be rewritten as $\frac{-B + -B + \sqrt{B^2 - 4AC} - \sqrt{B^2 - 4AC}}{2A}$, which is equivalent to $-\frac{2B}{2A}$, or $-\frac{B}{A}$. In the equation

$16x^2 - 4a + bx + \frac{ab}{4} = 0$, $A = 16$, $B = -4a + b$, and $C = \frac{ab}{4}$. Substituting 16 for A and $-4a + b$ for B in $-\frac{B}{A}$ yields $-\frac{-4a + b}{16}$, which can be rewritten as $\frac{1}{16}4a + b$. Thus, the sum of the solutions to the given equation is $\frac{1}{16}4a + b$. Since it's given that the sum of the solutions to the given equation is $k4a + b$, where k is a constant, it follows that $k = \frac{1}{16}$.

Q8**A****A. 23**

Choice A is correct. It's given that in the xy -plane, the graph of $y = g(x)$ passes through the point $(0, 22)$. It follows that $g(0) = 22$. Substituting 0 for x and 22 for $g(x)$ in the given equation yields $22 = \frac{0^2 - 0 - a}{0^3 - 0 - b}$, which is equivalent to $22 = \frac{-a}{-b}$, or $22 = \frac{a}{b}$. Multiplying each side of this equation by b yields $22b = a$. Substituting $22b$ for a in the equation defining the function g yields $gx = \frac{x^2 - x - 22b}{x^3 - x - b}$. It's given that $g(-22) = 0$. Substituting -22 for x and 0 for $g(x)$ in the equation $gx = \frac{x^2 - x - 22b}{x^3 - x - b}$ yields $0 = \frac{(-22)^2 - (-22) - 22b}{(-22)^3 - (-22) - b}$. Since a rational expression equals 0 if and only if its numerator is 0, it follows that $(-22)^2 - (-22) - 22b = 0$, or $22^2 + 22 - 22b = 0$. Dividing each side of this equation by 22 yields $22 + 1 - b = 0$, or $23 - b = 0$. Adding b to each side of this equation yields $23 = b$, so the value of b is 23.

Choice B is incorrect. This is the value of $g(0)$, not the value of b .

Choice C is incorrect. This is a value of x for which $g(x) = 0$, not the value of b .

Choice D is incorrect and may result from conceptual or calculation errors.

Q9**A**

A. $H = \frac{25}{9}(D - T) + 100$

Choice A is correct. It's given that $D = T - \frac{9}{25}(100 - H)$. Solving this formula for H expresses the relative humidity in terms of the temperature and the dew point. Subtracting T from both sides of this equation yields $D - T = -\frac{9}{25}(100 - H)$. Multiplying both sides by $-\frac{25}{9}$ yields $-\frac{25}{9}(D - T) = 100 - H$. Subtracting 100 from both sides yields $-\frac{25}{9}(D - T) - 100 = -H$. Multiplying both sides by -1 results in the formula $\frac{25}{9}(D - T) + 100 = H$.

Choices B, C, and D are incorrect and may result from errors made when rewriting the given formula.

Q10**D**

D. The time at which the projectile hits the ground

Choice D is correct. The positive x-intercept of the graph of $y = h(x)$ is a point (x, y) for which $y = 0$. Since $y = h(x)$ models the height above the ground, in feet, of the projectile, a y-value of 0 must correspond to the height of the projectile when it is 0 feet above ground or, in other words, when the projectile is on the ground. Since x represents the time since the projectile was launched, it follows that the positive x-intercept, $(x, 0)$, represents the time at which the projectile hits the ground.

Choice A is incorrect and may result from misidentifying the y-intercept as a positive x-intercept. Choice B is incorrect and may result from misidentifying the y-value of the vertex of the graph of the function as an x-intercept. Choice C is incorrect and may result from misidentifying the x-value of the vertex of the graph of the function as an x-intercept.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

Generated 2026-05-29 · cb-educator-question-bank